Percentage Practice Problems

1) A shopkeeper buys 120 shirts. He sells 70 of them at a profit of 15% and the remaining 50 shirts at a profit of 25%. If he had sold all the shirts at a uniform profit of 18%, he would have received ₹ 75 more. Find the cost price of each shirt.
2) A vendor purchases 200 pens. He sells 120 pens at a gain of 12% and the rest at a loss of 5%. If he had sold all the pens at a uniform profit of 7%, he would have gained ₹ 60 less. What is the cost price of one pen?
3) A fruit seller buys a certain number of apples. He sells 60% of them at a profit of 20% and the remaining at a profit of 10%. If his total profit was ₹ 320, and he had sold all the apples at a uniform profit of 16%, his total profit would have been ₹ 20 less. Find the total number of apples he purchased if the cost price of each apple is ₹ 10.
4) A toy manufacturer produces 300 toys. He sells one-third of them at a profit of 25% and the rest at a profit of 10%. If he had sold all the toys at a uniform profit, he would have realized 2% less profit on the total cost. Calculate the uniform profit percentage if the cost price of each toy is ₹ 50.

Type 1: Profit % = Loss %

5) The percentage profit earned by selling an article for ₹2500 is equal to the percentage loss incurred by selling the same article for ₹1500. At what price should the article be sold to make 20% profit?
6) If selling an item for ₹840 results in the same percentage profit as selling it for ₹600 results in percentage loss, what is the cost price of the item? What would be its selling price if a 10% profit is desired?
7) A shopkeeper finds that the profit percentage on selling a toy for ₹360 is the same as the loss percentage on selling it for ₹240. Find the price at which he should sell the toy to gain 15%.

Type 2: Ratio of Profit % to Loss %

8) The percentage profit earned by selling an article for ₹900 is twice the percentage loss incurred by selling the same article for ₹450. Find the cost price of the article.
9) When an article is sold for ₹1200, there is a certain percentage profit. When the same article is sold for ₹600, there is a percentage loss which is half of the percentage profit. What is the cost price of the article?

Type 3: Absolute Profit/Loss

10) Selling an article for ₹480 leads to a profit of 20%. If the article is sold for ₹400, what is the percentage loss or profit?
11) A bicycle is sold for ₹1800 at a loss of 10%. At what price should it be sold to gain 15%?

Type 4: Combined Scenarios

12) By selling an article for ₹120, a man gains ₹20. What is his profit percentage? If he wants to earn 30% profit, at what price should he sell the article?
13) A trader sold an item for ₹69 and incurred a loss of 8%. At what price should he have sold the item to have gained 10%?

Problem Set 1: Basic Mixing and Selling

14) A shopkeeper buys 10 kg of apples at ₹50/kg and 5 kg of oranges at ₹60/kg. He mixes them and sells the entire fruit mixture for ₹55/kg. What is his profit or loss percentage?
15) You have 15 liters of milk costing ₹40/liter and 10 liters of water costing ₹10/liter. You mix them together and sell the mixture at ₹35/liter. Calculate your profit or loss percentage.
16) A baker uses 5 kg of flour at ₹30/kg and 3 kg of sugar at ₹50/kg to make a cake mix. He sells the entire mix for ₹40/kg. What is his profit percentage?

Problem Set 2: Slightly More Complex Mixing

17) A grocer mixes 20 kg of pulses costing ₹80/kg with 30 kg of another variety of pulses costing ₹70/kg. He wants to make a 10% profit. At what price per kg should he sell the mixture?
18) You purchase 8 meters of cloth at ₹120/meter and 12 meters of another type of cloth at ₹100/meter. You mix them and sell the entire lot at a 5% loss. What is the selling price per meter of the mixed cloth?

Problem Set 3: Finding Unknowns (Slightly more advanced)

19) A tea seller mixes 5 kg of tea at ₹150/kg with another variety of tea. If he sells the mixture at ₹180/kg and makes a 20% profit, and the total quantity of the mixture is 10 kg, what was the cost per kg of the second variety of tea?
20) A fruit vendor mixes 10 kg of grapes costing ₹100/kg with an unknown quantity of apples costing ₹120/kg. He sells the entire mixture for ₹115/kg and makes a 10% profit. What was the quantity of apples mixed?

Answers:

1) ₹ 53.57.
2) ₹ 16.67.
3) 200
4) 13%
5) ₹2400
6) CP = ₹720; SP = ₹792
7) ₹345
8) ₹500
9) ₹800
10) Profit of 33.33% (or 33 1/3%)
11) ₹2300
12) Profit percentage = 20%; SP = ₹130
13) ₹82.50

Problem Set 1: Basic Mixing and Selling

14) Profit of 3.13%
15) Profit of 25%
16) Profit of 6.67%

Poblem Set 2: Slightly More Complex Mixing

17) ₹81.4/kg
18) ₹102.6/meter

Problem Set 3: Finding Unknowns

19) ₹150/kg
20) 2.94 kg